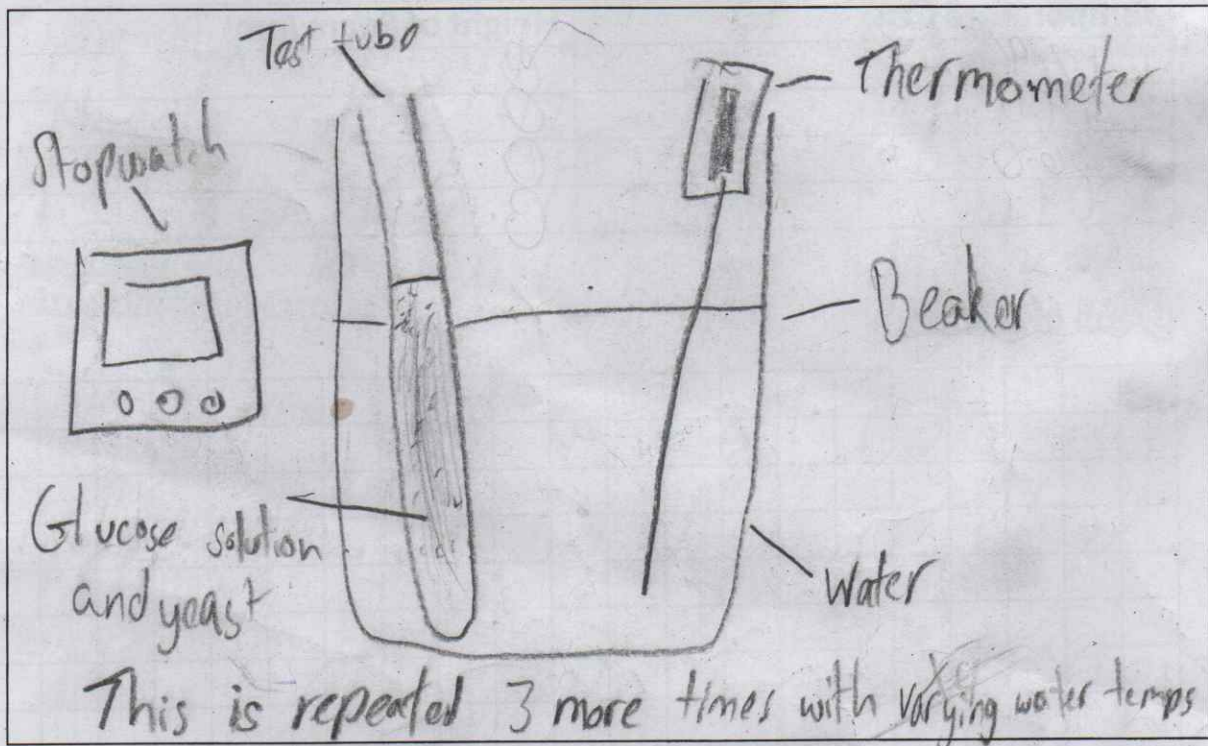


The Effect of Temperature on Respiration of Yeast

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Date of Experiment: 3/3/16

Labelled Diagram of equipment setup:



Method (Bullet Points):

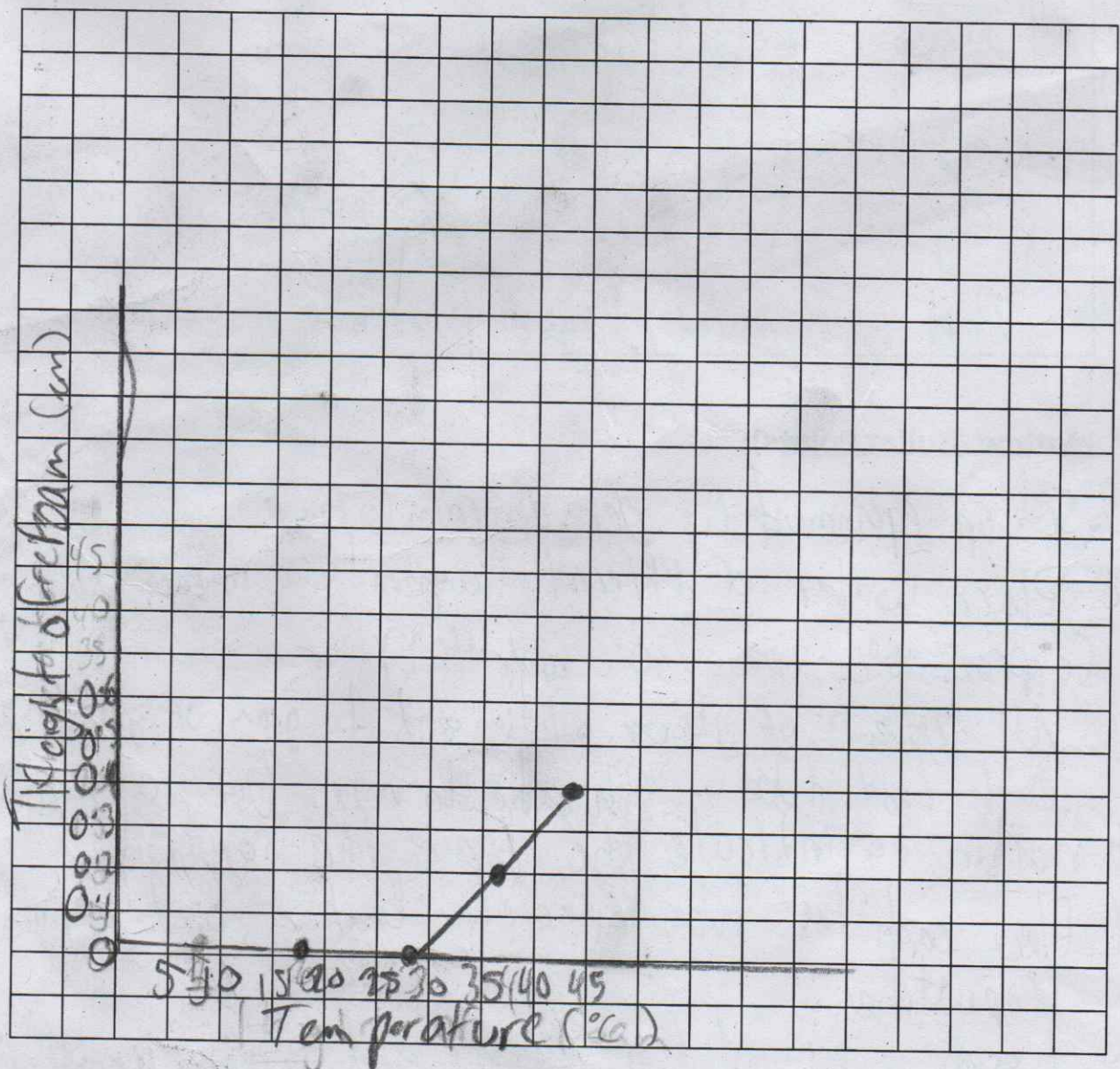
1. Set up equipment as per diagram above
2. Place water of different temperatures in each beaker (approx. 10°C, 20°C, 30°C and 40°C)
3. Add 20 cm³ of glucose solution and 1 gram of yeast to each test tube. ~~Shake~~ Shake to mix. Use a permanent marker to indicate the top of the solution.
4. Place each test tube in each beaker and begin the stopwatches
5. Leave experiment for 20 mins and measure height of foam

Independent Variable: Temperature
 Dependent Variable: Height of foam (carbon dioxide)
 Control: Yeast in water

Results:

Temperature (°C)	Height of Foam (cm)
1. 17.9	0
2. 29.4	0
3. 36.0	0.2
4. 44.1	3.1

Graph of results:



Analysis and Conclusion:

We found that higher temperatures led to more foam, suggesting yeast needs high temperatures to operate. At ten and twenty $^{\circ}\text{C}$, the temperature is too low for respiration to take place. At 40°C , the height of foam decreased. This is due to enzymes not being able to function at high temperatures.

Experimental Errors:

Over time the water got colder as it released its heat into the air. It is really difficult to keep the temperature constant.

f foam